

HARI KRISHNA KANCHARLA

Boston, MA | 508-293-1724 | harikrishnak2426@zohomail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Software Engineer with over 4 years building production LLM and agentic systems, from retrieval and fine-tuning to high-throughput serving. At Morgan Stanley, builds retrieval and inference platforms for analyst and trading teams, owning retrieval quality, evaluation, guardrails, and reliability at scale. Experienced across the full stack, from backend APIs to model fine-tuning and Kubernetes serving optimization. Strong in Python, Kotlin, and Java, with LangGraph, vector retrieval, and MLOps tooling.

SKILLS

Languages: Python, C++, SQL, TypeScript, JavaScript, Kotlin, Java

Frontend and Backend: React, Next.js, Node.js, Tailwind, Spring Boot, FastAPI, GraphQL, REST, gRPC, WebSockets, microservices, Kafka

Machine Learning and AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, LightGBM, LangChain, LangGraph, LangSmith, MCP, RAG, DDP, agent orchestration, tool calling, evals, LLM-as-judge

Model Optimization and Serving: LoRA, QLoRA, ONNX Runtime, TensorRT, vLLM, FlashAttention, 4-bit quantization, batching, caching

MLOps and Data: MLflow, Weights & Biases, Airflow, Kubeflow, drift detection, A/B testing, FAISS, Pinecone, Spark

Cloud and Infrastructure: GCP, AWS, Shell Scripting, Azure, Linux/Unix, Docker, Kubernetes, GitHub Actions, PostgreSQL, MongoDB, Redis

EXPERIENCE

Morgan Stanley, Boston, MA

Nov 2024 to Present

AI Software Engineer

- Built a RAG platform on LangChain, Pinecone, and FAISS with hybrid retrieval (vector plus BM25), metadata-aware chunking, cross-encoder reranking, and FinBERT financial embeddings, improving recall@10 by 42% and cutting analyst research time 35%.
- Built CI/CD and deployment automation with GitHub Actions, ArgoCD, Docker, and Kubernetes for model versioning, rollback, and reproducibility, plus Prometheus and Grafana monitoring tracking hallucination rate, retrieval quality, and drift in production.
- Designed Kotlin and Spring Boot services exposing GraphQL APIs consumed across multiple internal teams, with schema-first design, federated resolvers, and end-to-end TypeScript type generation for the React client.
- Developed LoRA-based fine-tuning pipelines for LLaMA and Mistral using PyTorch, Hugging Face, and MLflow, improving domain-specific response precision by 28% while reducing training cost by 40%.
- Reduced p95 inference latency by 3.5x by optimizing Kubernetes-hosted LLM services with ONNX Runtime, TensorRT, quantization, and batching, supporting 1M+ daily inference requests.
- Built LLM workflows and eval harnesses (LLM-as-judge, golden datasets) with monitoring across retrieval quality, hallucination risk, and latency, catching regressions and improving production reliability.

Infinite Infolab, India

May 2021 to Jul 2023

Machine Learning Engineer

- Developed supervised and ensemble forecasting models (XGBoost, LightGBM, Scikit-learn) across SKU behavioral clusters with time-series cross-validation and rolling-window backtesting, improving demand prediction accuracy by 27%.
- Built LSTM-based deep learning models in TensorFlow and PyTorch for highly seasonal SKUs, capturing long-term temporal dependencies and reducing forecast error (MAPE) by 22%.
- Engineered ETL and feature pipelines in Python and SQL processing 10M+ records from ERP, POS, and supplier systems, cutting preprocessing time 55% through parallelization, query tuning, and reusable feature modules.
- Deployed scalable inference APIs with FastAPI and Docker on GCP, supporting 50K+ daily prediction requests at 99.9% uptime via horizontal autoscaling, batch optimization, and feature caching.
- Automated experiment tracking, model versioning, and drift monitoring with MLflow, enabling safe production rollbacks and doubling experimentation efficiency across forecasting models.

PROJECTS

[Helm Browser Agent](#), **Agentic Browser Runtime** | Python, FastAPI, WebSockets, Playwright, async SQLite, TypeScript

- Built an autonomous browser-agent runtime that converts natural-language tasks into browser actions through a perceive-decide-act loop with page-state grounding, screenshots, bounded steps, and loop detection.
- Engineered a fault-tolerant LLM cascade across Groq Llama 3.3 70B, Gemini 2.0 Flash, and local Ollama, with quota-aware fallback, structured JSON action parsing, chained workflows, scheduling, session recording, and Playwright export.

[Code Review Arena](#), **AI Code Review Evaluation Platform** | Python, FastAPI, Typer, Pytest, Pydantic, Docker, Next.js

- Built an execution-backed benchmark for AI code-review agents that separates bug detection from validated repair, crediting fixes only when patches apply cleanly, pass regression tests, and satisfy structural validators inside sandboxed per-run workspaces.
- Designed a patch-required audit suite across security, distributed systems, RAG safety, concurrency, reliability, and API correctness, with baseline reviewers, CLI runs, leaderboard metrics, dashboard pages, and generated audit reports.

EDUCATION

M.S. in Data Analytics Engineering, Concentration: Machine Learning | **Northeastern University, Boston, MA**

May 2025